VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-09 PHYSICS

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Topics To Be Covered

1. A.C. GENERATOR
2. DISTINCTION BETWEEN AN A.C. GENERATOR AND D.C. MOTOR
3. TRANSFORMER
4. QUESTIONS


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Physics Wallah

Frequency of Alternating Current

$\mathcal{E} = \mathcal{E}_0 \sin 2\pi ft$

↓

EMF Amplitude

$\mathcal{E}_0 = NBA\omega$

$I = \frac{\mathcal{E}}{R} = \frac{\mathcal{E}_0 \sin 2\pi ft}{R}$

$I = I_0 \sin 2\pi ft$

$I_0 = \frac{NBA\omega}{R}$

In one complete rotation of coil, we get one cycle of alternating current in the external circuit. Thus the alternating current has the frequency equal to the frequency of rotation of coil.

If the coil makes n rotations per second, the magnitude of induced e.m.f. at any instant t is given as

$$e = e_0 \sin 2\pi nt$$

and the current is expressed as

$$i = i_0 \sin 2\pi nt$$

where e_0 and i_0 represent the maximum values of e.m.f. and current respectively.

The maximum value e_0 of e.m.f. induced can be increased

(1) by increasing the number of turns in coil,
(2) by increasing the area of cross section of coil, and
(3) by increasing the speed of rotation of coil.



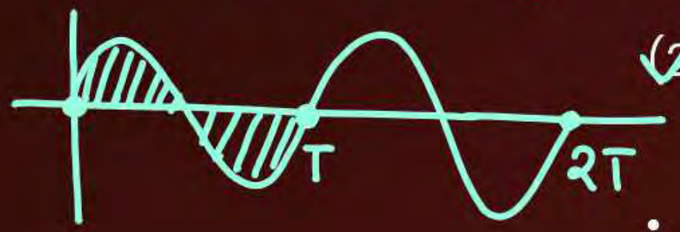
Note

$$\mathcal{E}_0 = NBA\omega$$

220V.

$f = 50\,Hz. = 50$ cycles in 1s

(1) By increasing the speed of rotation of coil, the frequency of a.c. generated will also increase, along with the increase in the maximum value e_0 of the induced e.m.f.



(2) The magnitude of alternating current changes continuously with time in a periodic manner.

In our houses, we have electric supply of frequency 50Hz i.e., the polarity at the supply terminal changes 100 times a second (50 times + and 50 times).

In USA, $f = 60\,Hz.$

$V = 110$ volts.

11000 V. ⟶ 132000 Volts. ⟶ 33000 V

V↑ I↓ H↓

$P = VI = \text{Const}$

$\boxed{I \propto \frac{1}{V}}$

33000 V

↓

11000 V

↓

220 V. (Home)

Difference between the d.c. and a.c.

Direct current (d.c.)	Alternating current (a.c.)
1. It is the current of constant magnitude.	1. It is the current of magnitude varying magnitude varying perodically with time.
2. If flows in one direction in the circuit.	2. It reverses its direction periodically while flowing in a circuit.
3. It is obtained from a cell (or battery)	3. It is obtained from an a.c. generator and mains.



Advantage of a.c. over d.c.

$V_{max} = 311$ Volts.

- In India, we use 220 volt a.c. in our houses and factories. The use of a.c. is advantageous over d.c. because only the voltage of a.c. can be increased by the use of step up transformer at the power generating station and then can be decreased by the use of step down transformer for transmission in the city.
- It reduces the loss of electrical energy as heat in the transmission line wires.
- If d.c. is generated at the power generating station, its voltage cannot be increased for transmission, and so due to passage of high current in the transmission line wires, there will be a huge loss of electrical energy as heat in the line wires.

DISTINCTION BETWEEN AN A.C. GENERATOR AND D.C. MOTOR

HW

A.C. generator	D.C. motor
1. A generator is a device which converts the mechanical energy into the electrical energy.	1. A d.c. motor is a device which converts the electrical energy into the mechanical energy.
2. A generator works on the principle of electromagnetic induction.	2. A d.c. motor works on the principle of force acting on a current carrying conductor placed in a magnetic field.
3. In a generator, the coil is rotated in a magnetic field so as to produce an electric current.	3. In a d.c. motor, the current from a d.c. source flows in the coil placed in a magnetic field due to which the coil rotates.
4. A generator makes use of two separate coaxial slip rings.	4. A d.c. motor makes use of two parts of a slip ring (i.e., split ring) which acts as a commutator.



SIMILARITIES BETWEEN AN A.C. GENERATOR AND D.C. MOTOR

(1) Both in an a.c. generator and d.c. motor, a coil rotates in a magnetic field between the pole pieces of a powerful electromagnet.

(2) Both in an a.c. generator and d.c. motor, there is a transformation of energy from one form to the other form.



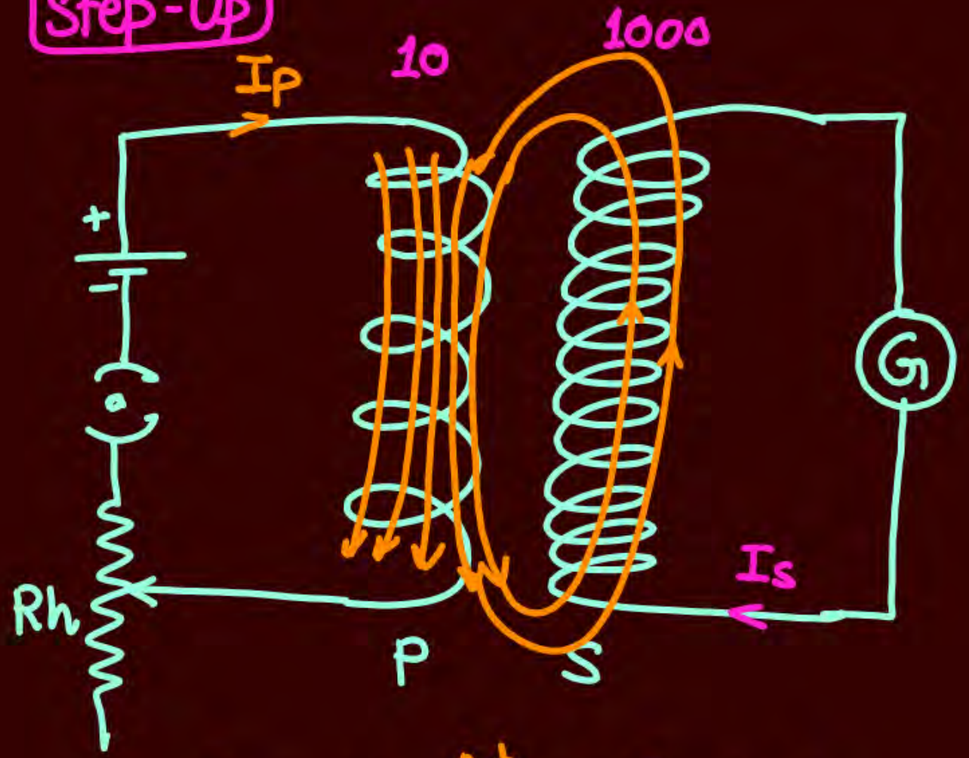
TRANSFORMER

- In our daily life, we use various electrical appliances which require working voltage different from the voltage of mains (i.e., 220 V) e.g. a door bell needs 6 V , while the cathode ray tube in a television needs nearly 10,000 V. To provide suitable voltage to different appliances from the mains, we use transformers with them.

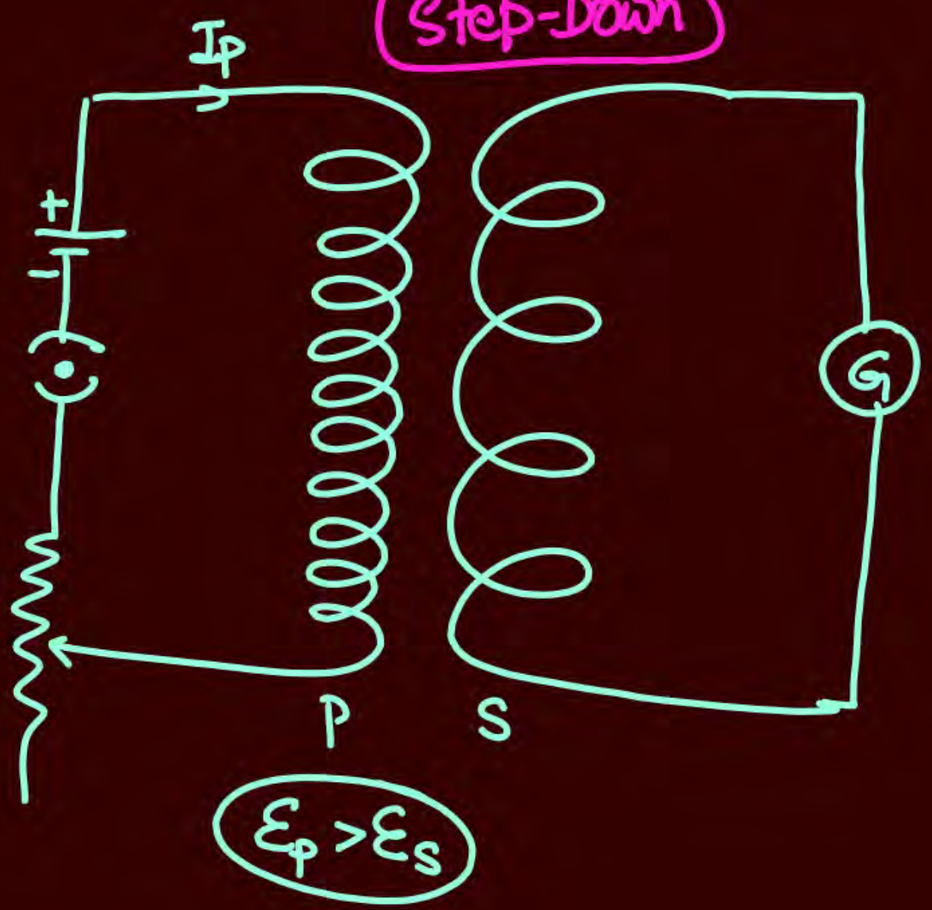
✓ Transformer is a device by which the magnitude of an alternating e.m.f. can be increased or decreased.

✓ A transformer does not affect the frequency of the alternating voltage. The frequency remains unchanged (=50 Hz).

Step-Up



Step-Down



$\varepsilon_p > \varepsilon_s$

$$\varepsilon_p = N_p \frac{\Delta\phi}{\Delta t} \Rightarrow \varepsilon_p \propto N_p$$

$$\varepsilon_s = N_s \frac{\Delta\phi}{\Delta t} \Rightarrow \varepsilon_s \propto N_s$$

$\varepsilon_s > \varepsilon_p$

In ideal case,

$$\frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p}$$

Turn Ratio



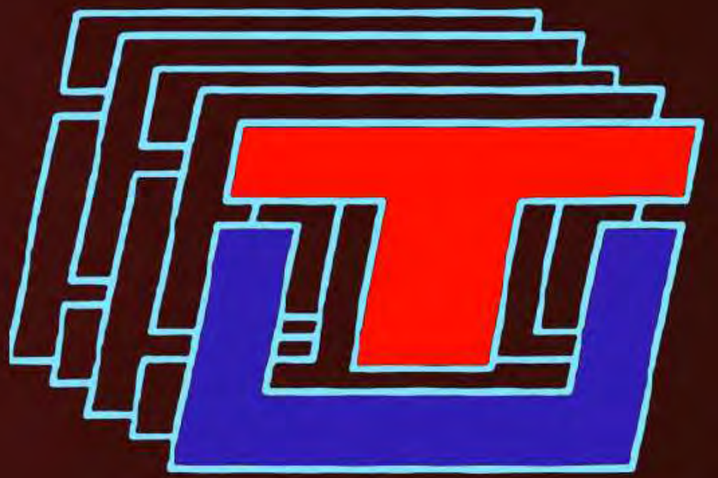
TRANSFORMER

Principle: A transformer works on the principle of electromagnetic induction and makes use of two coils having different number of turns. The two coils are : (1) the primary coil, and (2) the secondary coil.

- The given alternating e.m.f., of which magnitude is to be changed, is applied across the primary coil and the appliance in which output is to be obtained is connected across the secondary coil.
- When there is a change in the magnetic field lines due to varying current in the primary coil, the number of magnetic field lines linked with the secondary coil changes and so an induced varying current of same frequency, but of different magnitude flows in the secondary coil.
- A transformer cannot be used with the direct current (d.c.) source because if current in primary coil is constant (i.e., d.c.), then there will be no change in the number of magnetic field lines linked with the secondary coil and hence no e.m.f. will be induced in the secondary coil.

TRANSFORMER

- **Soft Iron Core**: A rectangular core is made of thin, laminated T- and U-shaped soft iron sheets insulated from each other by paint or varnish to reduce energy losses.



- **Primary Coil** (P): An insulated copper wire coil wound on one arm of the core, connected to the A.C. input source.
- **Secondary Coil** (S): An insulated copper wire coil wound on the other arm of the core, from which the output e.m.f. is obtained.
- **Turns Ratio**: The ratio of the number of turns in the secondary coil to that in the primary coil is called the turns ratio.

- **Turns ratio** $n = \dfrac{\text{number of turns in secondary coil } N_s}{\text{number of turns in primary coil } N_p}$



Note

(1) The laminated core prevents the loss of energy due to induced (or eddy) currents in the core.

(2) The advantage of using a closed core is that it gives a closed path for the magnetic field lines and therefore almost all the magnetic field lines, caused due to current in the primary coil, remain linked with the secondary coil (i.e., the flux linkage is nearly perfect) and loss of energy is avoided.

(3) The core is made of soft iron so that the loss of energy in the core gets minimised.



Working

- When the terminals of primary coil are connected to the source of alternating e.m.f., a varying current flows through the primary coil.
- This varying current produces a varying magnetic field in the core of transformer. Thus the number of magnetic field lines linked with the secondary coil vary due to which an e.m.f. induces in it.
- The induced e.m.f. varies in the same manner as the applied e.m.f. in the primary coil varies and thus has the same frequency as that of the applied e.m.f.

$$\frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p} = n$$

$$\frac{\text{e.m.f. across the secondary coil}}{\text{e.m.f. across the primary coil}} = \frac{\text{number of turns in the secondary coil } (N_S)}{\text{number of turns in the primary coil } (N_p)}$$

Thus e.m.f. induced in the secondary coil depends on the following two factors:

(1) turns ratio,

(2) the magnitude of e.m.f. applied in the primary coil.

$$\varepsilon_s = \frac{N_s}{N_p}\,\varepsilon_p$$

Working

For an ideal transformer, when there is no energy loss, the output power will be equal to the input power.

$$P_p = P_s$$

$$\mathcal{E}_p I_p = \mathcal{E}_s I_s$$

$$\frac{\mathcal{E}_s}{\mathcal{E}_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$$

power in secondary coil = power in primary coil or

$$E_s I_s = E_p I_p$$

But

$$\frac{I_p}{I_s} = \frac{E_S}{E_p} = \frac{N_s}{N_p}$$



Types of Transformers

There are two types of transformers:

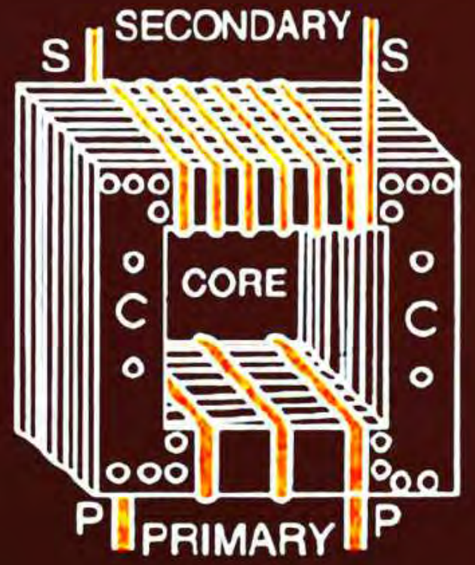
(1) step up transformer, and

(2) step down transformer

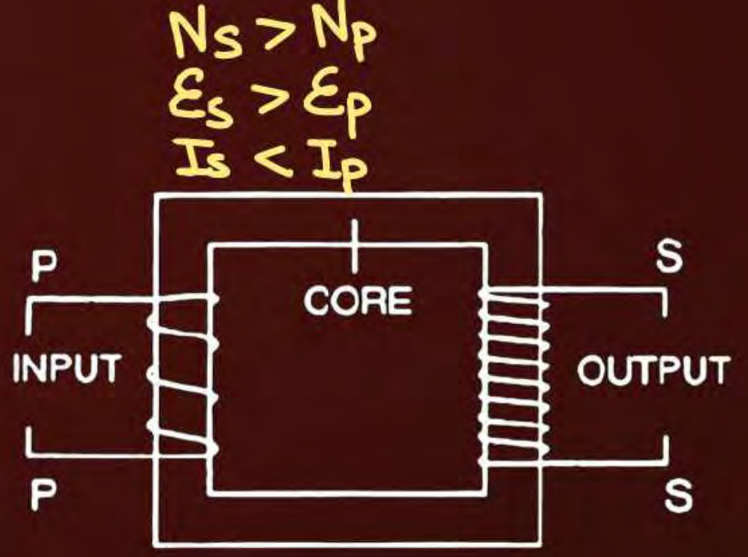
Types of Transformers

Step up transformer: The transformer used to change a low alternating voltage to a high alternating voltage (of same frequency) is called a step up transformer (i.e., $E_S > E_P$).

- In a step up transformer, $(N_s > N_p)$ i.e., turns ratio $n > 1$.
- Since current in primary coil is more than in secondary coil (or $I_P > I_S$), so the wire in the primary coil is thicker than in the secondary coil.



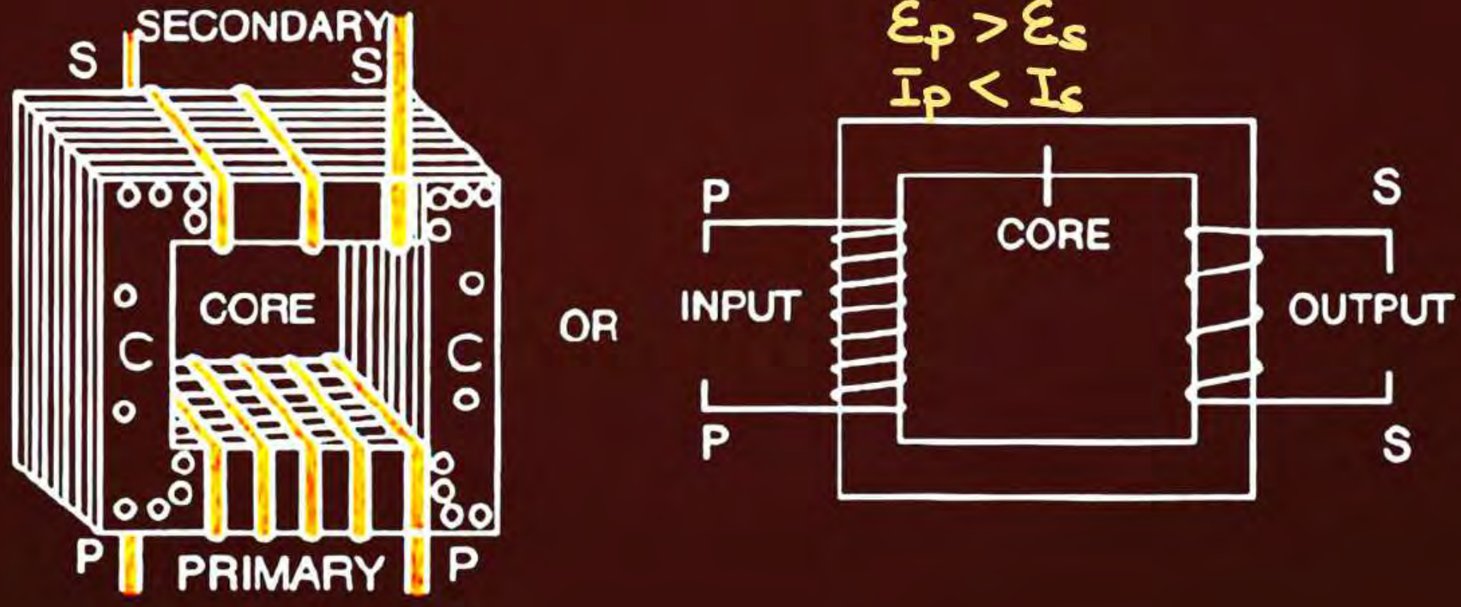
OR



Types of Transformers

Step down transformer: The transformer used to change a high alternating voltage to a low alternating voltage (of same frequency) is called a step down transformer ($E_S < E_P$).

- In a step down transformer, ($N_s < N_p$) i.e., turns ratio $n < 1$.
- Since current in secondary coil is more than in the primary coil (i.e., $I_S > I_p$) so the wire in secondary coil is thicker than in the primary coil.





Uses of Transformers

(A) USES OF STEP UP TRANSFORMERS :

(1) In transmission of electric power at the power generating station to step up the voltage.

(2) With television.

(3) With wireless sets.

(4) With X-ray tubes to provide a high accelerating voltage.

(B) USES OF STEP DOWN TRANSFORMERS :

(1) With electric bells, night electric bulbs, mobile phones, computers, etc.

(2) At the power sub-stations to step down the voltage before its distribution to the consumers.

Distinction between the step up and step down transformers

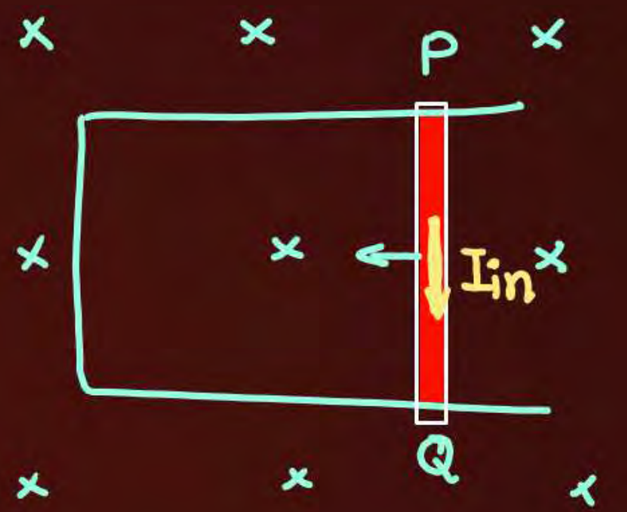
HW

Step up transformer	Step down transformer
1. It increases the a.c. voltage and decreases the current. i.e., $E_s > E_p$ and $I_s < I_p$.	1. It decreases the a.c. voltage and increases the current. i.e., $E_s < E_p$ and $I_s > I_p$.
2. The turns ratio $N_s/N_p > 1$ i.e., it has more number of turns in the secondary coil than in the primary coil.	2. Its turns ratio $N_s/N_p < 1$ i.e., it has less number of turns in the secondary coil than in the primary coil.
3. The wire of primary coil is thicker than that of the secondary coil.	3. The wire of the secondary coil is thicker than that of the primary coil.
Uses: At power generating station, with X-ray tubes, television, etc.	**Uses:** At power substations, with night electric bulbs, computers, electric bells, etc.

QUESTIONS

The direction of induced current is obtained by:

1. Fleming's left hand rule
2. Clock rule
3. Right hand thumb rule
4. Fleming's right hand rule.



$I_{induced}$?

(a) P to Q

(b) Q to P



QUESTIONS

In a step up transformer

1. $N_S = N_P$
2. $N_S < N_P$
3. $N_S > N_P$
4. Nothing can be said

QUESTIONS

A coil has 50 turns and the magnetic flux linked with the coil increases by 0.5 weber in 10.0 s. Calculate the e.m.f. induced in the coil.

$N = 50$

$\Delta\phi = 0.5$ Wb

$t = 10$ s.

$$\varepsilon_{in} = \frac{N\Delta\phi}{\Delta t}$$

$$= \frac{50 \times 0.5}{10}$$

$= 2.5$ V.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

Fig. shows a coil AB connected to a centre zero galvanometer G and a magnet NS.

(i) What will you observe when :

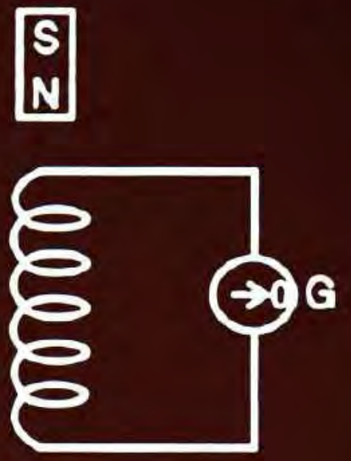
(a) the magnet is inserted into the coil,

(b) the magnet is taken out of the coil ?

(ii) How will your observation in part (i) change if the number of turns in the coil is increased?

(iii) State the direction of current in the coil in part (i) (a).

(iv) State the law by which you arrived to the answer in part (iii).



(i) (a) As the magnet enters the coil, a momentary deflection is observed in the galvanometer so long the magnetic field lines linked with the coil increases. As the magnet completely enters inside the coil, the deflection becomes zero.

(b) On taking the magnet out of the coil, again a momentary deflection is observed in the galvanometer in a direction opposite to that in part (a) above. The deflection lasts so long as the magnetic field lines linked with the coil decreases.

(ii) On increasing the number of turns in the coil, the deflection observed in part (i) will increase.

(iii) In part (i) (a), the current in coil will be from A to B (i.e. anticlockwise as seen from above, so that the end of the coil near the N pole of the magnet becomes the north pole).

(iv) According to Lenz's law, the current induced in the coil is such that it opposes the cause of its origin.

QUESTIONS

Two coils P and Q are placed as shown in Fig. 10.39. The coil A is connected to a battery B and a key K, while the coil Q is connected to a centre zero galvanometer G.What will you observe in the galvanometer G when

(i) the key K is closed?

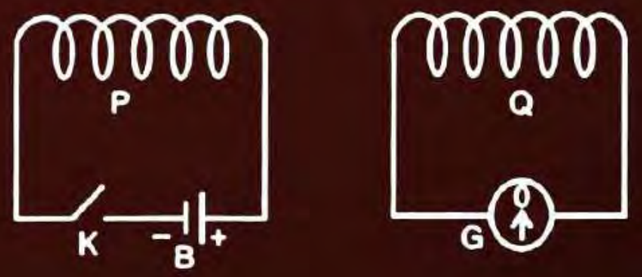
(ii) the key K is opened?

(iii) with the key K closed, the coil P is moved rapidly towards the coil Q?

(iv) with the key K closed, the coil Q is moved rapidly towards the coil P?

(v) with the key K closed, the coils P and Q are moved away from each other?

(vi) with the key K closed, the coils P and Q are moved with the same speed in same direction?



(i) As the key K is closed, the galvanometer needle deflects for a moment and then returns to zero.

(ii) As the key K is opened, again a momentary deflection (but more) is observed in the galvanometer in the opposite direction.

(iii) With the key K closed, if the coil P is moved rapidly towards the coil Q , a deflection is obtained in the galvanometer in the direction same as in case (i). But the deflection lasts so long as the coil P moves.

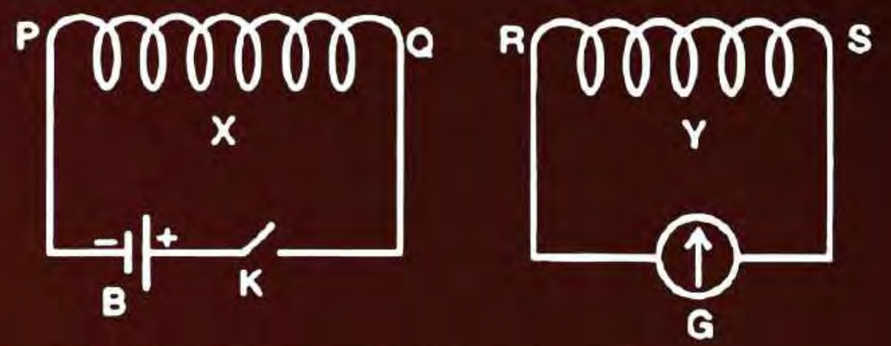
(iv) With the key K closed, if the coil Q is moved rapidly towards the coil P , again a deflection is observed in the galvanometer in direction same as in case (i). The deflection lasts so long as the coil Q moves.

(v) With the key K closed, if the coils P and Q are moved away from each other, a deflection is obtained in the galvanometer in direction same as in case (ii). The deflection lasts so long as there is relative motion between the coils.

(vi) With the key K closed, if the coils P and Q are moved with the same speed in the same direction, there is no deflection in the galvanometer.

QUESTIONS

The diagram in Fig. 10.40 shows two coils X and Y. The coil X is connected to a battery B and a key K . The coil Y is connected to a galvanometer G .
When the key K is closed, state the polarity
(i) at the end Q of the coil X ,
(ii) at the end R of the coil Y ,
(iii) at the end R of the coil Y if the coil Y is
(a) moved towards the coil X , (b) moved away from the coil X.



(i) On closing the key K, the current at the end Q of the coil X is anticlockwise, therefore at this end there is a north pole.

(ii) While closing the key, polarity at the end R of the coil Y will be north. But there will be no polarity at the end R of the coil Y when the current becomes steady in the coil X after the key has been closed.

(iii) (a) With the key K closed, while the coil Y is moved towards the coil X , the polarity at the end R of the coil Y is north.

(b) With the key K closed, while the coil Y is moved away from the coil X, the polarity at the end R of the coil Y is south.

Thank You